AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

Department of Computer Science and Engineering

Course No: CSE 2104
Course Title: Data Structures Lab

Assignment No: 06

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CSE2104 Offline-6 Report

Problem Statement / Task:

Write a program to support multi-digit numbers in the conversion of an infix expression to postfix notation and the evaluation of the resulting postfix expression.

For example, your code should convert this infix expression to its appropriate postfix notation: **(12 + 34) * 56 - 78 / 9**.

Then evaluate the result of this expression from the postfix notation.

Source Code (C++):

```
#include<bits/stdc++.h>
using namespace std;

stack<char> st;
stack<int> it;

int piority(char c)
{
    if(c == '^') return 3;
    else if(c == '*' || c == '/') return 2;
    else if(c == '+' || c == '-') return 1;
    else return -1;
}

string infixToPostfix(string s)
{
    string postfix = "";

    for(int i = 0; i < s.length(); i++)
    {
        if(s[i] >= '0' && s[i] <= '9')
        {
            while(i < s.length() && s[i] >= '0' && s[i] <= '9')
            {
                postfix += s[i];
                i++;
            }

            postfix += ' ';
            i--;
        }

        else if(s[i] == '(')
        {
            st.push(s[i]);
        }
```

```
        else if(s[i] == ')')
        {
            while(!st.empty() && st.top() != '(')
            {
                postfix += st.top();
                postfix += ' ';
                st.pop();
            }

            if(!st.empty())
                st.pop();
        }

        else
        {
            while(!st.empty() &&
                  st.top() != '(' &&
                  piority(s[i]) <= piority(st.top()))
            {
                postfix += st.top();
                postfix += ' ';
                st.pop();
            }

            st.push(s[i]);
        }
    }

    while(!st.empty())
    {
        postfix += st.top();
        postfix += ' ';
        st.pop();
    }

    return postfix;
}

int evaluation(string post)
{
    for(int i = 0; i < post.size(); i++)
    {
        if(post[i] >= '0' && post[i] <= '9')
        {
            int n = 0;

            while(i < post.size() &&
                  post[i] >= '0' && post[i] <= '9')
```

```
        {
            n = n * 10 + (post[i] - '0');
            i++;
        }

        it.push(n);
        i--;
    }

    else if(post[i] == '+')
    {
        int a = it.top();
        it.pop();

        int b = it.top();
        it.pop();

        it.push(b + a);
    }

    else if(post[i] == '-')
    {
        int a = it.top();
        it.pop();

        int b = it.top();
        it.pop();

        it.push(b - a);
    }

    else if(post[i] == '*')
    {
        int a = it.top();
        it.pop();

        int b = it.top();
        it.pop();

        it.push(b * a);
    }

    else if(post[i] == '/')
    {
        int a = it.top();
        it.pop();

        int b = it.top();
```

```
            it.pop();

            it.push(b / a);
        }
    }

    return it.top();
}

int main()
{
    cout << "Enter the infix expression: ";

    string s;
    cin >> s;

    string postfix = infixToPostfix(s);

    cout << "Postfix expression: " << postfix << endl;

    int ev = evaluation(postfix);

    cout << "Evaluation result : " << ev;

    return 0;
}
```

Execution Trace (Input & Output):

Console Output Screenshot:

```
Enter the infix expression: (12+34)*56-78/9
Postfix expression: 12 34 + 56 * 78 9 / -
Evaluation result : 2568
Process returned 0 (0x0)   execution time : 10.540 s
Press any key to continue.
```

<u>Execution Output Details:</u>

Enter the infix expression: (12+34)*56-78/9

Postfix expression: 12 34 + 56 * 78 9 / -

Evaluation result : 2568